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Moving Extreme Ranked Set Sampling for Imbalanced Binary Classification: A Logistic Regression Perspective

Abstract:

Logistic regression (LR) is widely used for binary classification, yet its performance is often compromised by class imbalance and inefficient sampling. Moving extreme ranked set sampling (MERSS) offers a potential solution by utilizing auxiliary information to target extreme observations where rare events reside, but its impact on LR classification has not been systematically evaluated. This study compares SRS and MERSS for LR classification under imbalanced conditions ($r = 0.2, 0.3, 0.4$), varying sample sizes ($n = 120, 210, 300$), correlations ($\rho = 0.3, 0.5, 0.7$), and set sizes ($k = 3, 5, 10$). Results demonstrate that MERSS consistently outperforms SRS, particularly for minority class identification. The largest gains were observed under the most severe imbalance ($r = 0.2$), where MERSS improved Precision by up to 15% and Minor-F1 by up to 29% compared to SRS, with no additional computational cost. These findings establish MERSS as a highly efficient sampling strategy for LR classification in imbalanced data, improving rare event prediction without increasing resource expenditure.

Keywords: Logistic regression, Moving extreme ranked set sampling (MERSS), Imbalanced classification

Mathematics Subject Classification (2010): 62D05, 62J12, 62P10.

1 Introduction

Logistic regression (LR) is widely used for binary classification due to its interpretability and computational efficiency. However, its predictive performance can be substantially degraded by class imbalance and inefficient sampling strategies, particularly when the minority class is rare [Hosmer et al. \(2013\)](#). Simple

random sampling (SRS), despite its widespread use, often fails to adequately represent minority classes under imbalance, especially with limited sample sizes [Noor et al. \(2022\)](#).

Rank-based sampling (RBS) methods offer a promising alternative by utilizing auxiliary information to select more informative samples without increasing sample size [McIntyre \(1952\)](#). Among these, moving extreme ranked set sampling (MERSS) [Al-Odat and Al-Saleh \(2001\)](#) is particularly suitable for imbalanced populations due to its focus on extreme observations, where rare events commonly reside. While RBS has demonstrated superior estimation efficiency compared to SRS [Samawi et al. \(2017, 2020\)](#), research linking these methods to LR classification remains scarce [Santos and Barrios \(2017\)](#); [Yousefi et al. \(2026\)](#). No study has systematically compared MERSS and SRS for LR classification under controlled imbalance. This study addresses this gap by evaluating both designs across varying imbalance ratios, sample sizes, correlations, and set sizes, emphasizing minority class identification.

2 Materials and Methods

2.1 Sampling Designs

Two sampling designs were compared. SRS selects units independently and with equal probability from the population. MERSS proceeds as follows. First, sets of sizes $1, 2, \dots, k$ are randomly drawn from the population. Within each set, units are ranked based on an auxiliary variable Z without measuring the response Y . From the set of size i , the extreme observation (maximum or minimum) is selected for measurement. This process constitutes one cycle, and m cycles are repeated to obtain a sample of size $n = m \times k$. By reviewing $\frac{k(k+1)}{2}$ units but measuring only k per cycle, MERSS efficiently targets distribution tails where minority classes often reside.

2.2 Logistic Regression Model

For a binary outcome $Y \in \{0, 1\}$ and predictor vector $\mathbf{X}$, the logistic regression model specifies

$$P(Y = 1 \mid \mathbf{X}) = \frac{\exp(\mathbf{X}^\top \boldsymbol{\beta})}{1 + \exp(\mathbf{X}^\top \boldsymbol{\beta})},$$

where $\boldsymbol{\beta}$ is estimated via maximum likelihood. Under MERSS, the likelihood function accounts for the ranking structure. For a sample obtained via MERSS, the log-likelihood is

$$l(\boldsymbol{\beta}) = \sum_{j=1}^m \sum_{i=1}^k [y_{[i]j} \log(\pi_{[i]j}) + (1 - y_{[i]j}) \log(1 - \pi_{[i]j})],$$

where $\pi_{[i]j}$ denotes the success probability for the extreme unit selected from the i -th set in the j -th cycle. Maximization is performed numerically using the Newton-Raphson algorithm.

2.3 Simulation Framework

Data were generated using a custom function adapted from the `TwoClassSim` function in the `caret` package in R. This function constructs a linear predictor comprising main effects, interactions, nonlinear terms, and noise variables. A key predictor Z , independent of other covariates, was included with tunable coefficient δ to control its association with the outcome. The binary response Y was generated via the inverse logit transformation. Three imbalance ratios were considered: $r = 0.2, 0.3, 0.4$. Three levels of correlation between Z and Y were set: $\rho = 0.3, 0.5, 0.7$. Sample sizes were $n = 120, 210, 300$. For MERSS, set sizes were $k = 3, 5, 10$. Each scenario was replicated $M = 200$ times.

2.4 Evaluation Metrics

Model performance was assessed using Accuracy, Precision, Precision-Recall AUC (Pr-AUC), Matthews Correlation Coefficient (MCC), Minor-F1, and Geometric Mean (G-Mean). Runtime was also recorded. Results are reported as mean (standard deviation) across replications.

3 Results

Tables 1, 2, and 3 present the performance of logistic regression models under imbalanced class conditions, comparing SRS with MERSS across varying sample sizes, correlation levels, and set sizes. Figures 1a, 1b, and 1c further illustrate the Minor-F1 performance for $k = 3, 5, 10$ at $\rho = 0.5$ across different sample sizes.

Overall Performance. MERSS consistently outperformed SRS across nearly all metrics and settings, with the largest gains observed under the most severe imbalance ratio ($r = 0.2$). The advantages were most pronounced for Precision, Minor-F1, and G-Mean, while Accuracy remained comparable between methods. These improvements were achieved with virtually no increase in computational cost.

Effect of Set Size (k). Increasing k from 3 to 10 monotonically improved MERSS performance, particularly for Precision and Minor-F1 (see Figures 1a–1c). For example, at $r = 0.2$, $n = 120$, and $\rho = 0.5$, Minor-F1 increased from 0.741 ($k = 3$) to 0.838 ($k = 10$), while SRS yielded only 0.659. Diminishing returns were observed at very high correlations ($\rho = 0.7$).

Effect of Imbalance Ratio (r). As imbalance became more severe (decreasing r from 0.4 to 0.2), the relative advantage of MERSS over SRS increased. At $r = 0.2$, SRS struggled substantially with minority class identification (e.g., Precision = 0.657 at $n = 120$, $\rho = 0.3$), whereas MERSS maintained robust performance (Precision = 0.754 at $k = 10$).

Effect of Correlation (ρ) and Sample Size (n). Higher ρ and larger n improved all metrics for both

Table 1: Performance metrics for logistic regression models under Imbalanced class condition ($r = 0.4$), evaluated across different sample sizes, varying set sizes, and correlation levels.

n	ρ	Method	k	Accuracy	Precision	Pr-AUC	MCC	Minor-F1	G-Mean	Run-Time
120	0.3	SRS	–	0.818 (0.052)	0.783 (0.783)	0.885 (0.047)	0.620 (0.108)	0.763 (0.072)	0.802 (0.058)	0.013 (0.002)
			3	0.808 (0.055)	0.804 (0.804)	0.890 (0.043)	0.617 (0.110)	0.798 (0.062)	0.806 (0.056)	0.013 (0.002)
		MERSS	5	0.820 (0.054)	0.824 (0.824)	0.904 (0.043)	0.641 (0.109)	0.824 (0.057)	0.817 (0.057)	0.012 (0.002)
			10	0.828 (0.051)	0.845 (0.845)	0.913 (0.039)	0.649 (0.106)	0.849 (0.047)	0.820 (0.056)	0.012 (0.002)
	0.5	SRS	–	0.837 (0.051)	0.809 (0.809)	0.908 (0.044)	0.660 (0.108)	0.788 (0.074)	0.823 (0.059)	0.017 (0.003)
			3	0.843 (0.051)	0.849 (0.849)	0.923 (0.037)	0.687 (0.103)	0.848 (0.052)	0.841 (0.055)	0.017 (0.003)
		MERSS	5	0.848 (0.051)	0.873 (0.873)	0.928 (0.034)	0.687 (0.104)	0.871 (0.046)	0.840 (0.056)	0.017 (0.003)
			10	0.865 (0.046)	0.891 (0.891)	0.937 (0.036)	0.686 (0.106)	0.900 (0.036)	0.832 (0.066)	0.017 (0.004)
	0.7	SRS	–	0.889 (0.042)	0.867 (0.867)	0.950 (0.032)	0.771 (0.085)	0.859 (0.056)	0.882 (0.045)	0.017 (0.003)
			3	0.892 (0.042)	0.905 (0.905)	0.956 (0.032)	0.781 (0.086)	0.905 (0.038)	0.888 (0.046)	0.017 (0.003)
		MERSS	5	0.901 (0.043)	0.928 (0.928)	0.952 (0.039)	0.771 (0.098)	0.927 (0.033)	0.882 (0.056)	0.017 (0.003)
			10	0.923 (0.034)	0.952 (0.952)	0.937 (0.058)	0.759 (0.113)	0.952 (0.022)	0.868 (0.083)	0.018 (0.003)
210	0.3	SRS	–	0.821 (0.040)	0.792 (0.792)	0.887 (0.035)	0.626 (0.081)	0.767 (0.053)	0.805 (0.043)	0.014 (0.002)
			3	0.823 (0.038)	0.818 (0.818)	0.905 (0.030)	0.646 (0.075)	0.813 (0.043)	0.821 (0.039)	0.013 (0.003)
		MERSS	5	0.824 (0.038)	0.825 (0.825)	0.909 (0.030)	0.649 (0.074)	0.828 (0.040)	0.823 (0.038)	0.013 (0.002)
			10	0.838 (0.042)	0.857 (0.857)	0.921 (0.029)	0.670 (0.085)	0.859 (0.039)	0.832 (0.045)	0.013 (0.002)
	0.5	SRS	–	0.843 (0.035)	0.813 (0.813)	0.916 (0.029)	0.673 (0.072)	0.799 (0.047)	0.832 (0.038)	0.018 (0.004)
			3	0.851 (0.033)	0.858 (0.858)	0.933 (0.023)	0.702 (0.065)	0.855 (0.033)	0.849 (0.034)	0.018 (0.005)
		MERSS	5	0.850 (0.034)	0.865 (0.865)	0.935 (0.023)	0.692 (0.071)	0.873 (0.031)	0.842 (0.038)	0.018 (0.004)
			10	0.875 (0.035)	0.903 (0.903)	0.946 (0.022)	0.709 (0.082)	0.909 (0.026)	0.847 (0.049)	0.018 (0.004)
	0.7	SRS	–	0.905 (0.030)	0.885 (0.885)	0.964 (0.018)	0.803 (0.063)	0.881 (0.040)	0.900 (0.032)	0.019 (0.004)
			3	0.903 (0.030)	0.912 (0.912)	0.968 (0.016)	0.803 (0.062)	0.916 (0.028)	0.900 (0.032)	0.020 (0.005)
		MERSS	5	0.916 (0.031)	0.937 (0.937)	0.973 (0.015)	0.806 (0.074)	0.939 (0.023)	0.900 (0.040)	0.019 (0.004)
			10	0.935 (0.025)	0.956 (0.956)	0.972 (0.026)	0.788 (0.083)	0.960 (0.016)	0.884 (0.058)	0.020 (0.004)
300	0.3	SRS	–	0.826 (0.033)	0.798 (0.798)	0.891 (0.030)	0.634 (0.069)	0.773 (0.046)	0.810 (0.036)	0.015 (0.002)
			3	0.822 (0.033)	0.816 (0.816)	0.903 (0.026)	0.644 (0.066)	0.811 (0.037)	0.821 (0.033)	0.014 (0.003)
		MERSS	5	0.827 (0.031)	0.831 (0.831)	0.911 (0.023)	0.655 (0.060)	0.832 (0.031)	0.826 (0.031)	0.014 (0.003)
			10	0.834 (0.031)	0.846 (0.846)	0.919 (0.020)	0.661 (0.062)	0.856 (0.029)	0.827 (0.033)	0.014 (0.003)
	0.5	SRS	–	0.848 (0.032)	0.821 (0.821)	0.920 (0.025)	0.682 (0.066)	0.806 (0.043)	0.837 (0.035)	0.018 (0.003)
			3	0.852 (0.028)	0.857 (0.857)	0.935 (0.019)	0.703 (0.056)	0.858 (0.029)	0.850 (0.028)	0.019 (0.004)
		MERSS	5	0.860 (0.028)	0.876 (0.876)	0.940 (0.018)	0.710 (0.060)	0.881 (0.025)	0.852 (0.032)	0.018 (0.003)
			10	0.874 (0.027)	0.900 (0.900)	0.949 (0.017)	0.708 (0.064)	0.909 (0.021)	0.844 (0.040)	0.018 (0.003)
	0.7	SRS	–	0.910 (0.022)	0.897 (0.897)	0.964 (0.015)	0.812 (0.047)	0.885 (0.031)	0.903 (0.025)	0.020 (0.003)
			3	0.909 (0.023)	0.920 (0.920)	0.971 (0.012)	0.814 (0.047)	0.921 (0.021)	0.906 (0.025)	0.021 (0.019)
		MERSS	5	0.919 (0.024)	0.934 (0.934)	0.975 (0.011)	0.813 (0.056)	0.940 (0.018)	0.902 (0.033)	0.020 (0.004)
			10	0.941 (0.020)	0.962 (0.962)	0.982 (0.010)	0.813 (0.063)	0.963 (0.013)	0.900 (0.043)	0.021 (0.003)

Note: Results are reported as Mean(SD) over $M = 200$ replications.

methods. MERSS consistently matched or exceeded SRS across all ρ levels (0.3, 0.5, 0.7) and all sample sizes ($n = 120, 210, 300$), with the clearest separation observed at moderate correlations ($\rho = 0.5$), as

Table 2: Performance metrics for logistic regression models under Imbalanced class condition ($r = 0.3$), evaluated across different sample sizes, varying set sizes, and correlation levels.

n	ρ	Method	k	Accuracy	Precision	Pr-AUC	MCC	Minor-F1	G-Mean	Run-Time
120	0.3	SRS	–	0.815 (0.055)	0.712 (0.712)	0.848 (0.065)	0.543 (0.135)	0.659 (0.110)	0.746 (0.087)	0.015 (0.012)
			3	0.809 (0.053)	0.756 (0.756)	0.872 (0.051)	0.589 (0.116)	0.731 (0.082)	0.784 (0.064)	0.012 (0.002)
		MERSS	5	0.802 (0.057)	0.770 (0.770)	0.875 (0.052)	0.593 (0.118)	0.755 (0.077)	0.791 (0.062)	0.013 (0.003)
			10	0.810 (0.052)	0.795 (0.795)	0.894 (0.042)	0.621 (0.102)	0.795 (0.056)	0.808 (0.052)	0.012 (0.002)
	0.5	SRS	–	0.853 (0.050)	0.756 (0.756)	0.902 (0.052)	0.645 (0.118)	0.740 (0.091)	0.813 (0.072)	0.016 (0.003)
			3	0.840 (0.046)	0.817 (0.817)	0.913 (0.038)	0.672 (0.096)	0.804 (0.062)	0.831 (0.051)	0.016 (0.003)
		MERSS	5	0.846 (0.054)	0.848 (0.848)	0.923 (0.039)	0.694 (0.109)	0.841 (0.059)	0.845 (0.056)	0.016 (0.003)
			10	0.853 (0.046)	0.875 (0.875)	0.929 (0.034)	0.692 (0.100)	0.878 (0.041)	0.840 (0.056)	0.017 (0.017)
	0.7	SRS	–	0.910 (0.039)	0.850 (0.850)	0.948 (0.042)	0.787 (0.091)	0.847 (0.069)	0.891 (0.054)	0.018 (0.003)
			3	0.901 (0.041)	0.892 (0.892)	0.953 (0.037)	0.803 (0.081)	0.893 (0.047)	0.900 (0.042)	0.017 (0.003)
		MERSS	5	0.903 (0.041)	0.910 (0.910)	0.955 (0.035)	0.801 (0.084)	0.916 (0.037)	0.898 (0.046)	0.018 (0.003)
			10	0.916 (0.040)	0.938 (0.938)	0.948 (0.049)	0.791 (0.097)	0.942 (0.030)	0.890 (0.059)	0.018 (0.003)
210	0.3	SRS	–	0.826 (0.037)	0.747 (0.747)	0.861 (0.044)	0.575 (0.090)	0.687 (0.069)	0.762 (0.050)	0.014 (0.003)
			3	0.822 (0.040)	0.780 (0.780)	0.881 (0.043)	0.617 (0.086)	0.752 (0.062)	0.799 (0.049)	0.013 (0.002)
		MERSS	5	0.820 (0.036)	0.789 (0.789)	0.889 (0.033)	0.628 (0.075)	0.775 (0.049)	0.809 (0.039)	0.013 (0.003)
			10	0.822 (0.037)	0.816 (0.816)	0.904 (0.029)	0.645 (0.073)	0.809 (0.040)	0.820 (0.037)	0.013 (0.003)
	0.5	SRS	–	0.864 (0.035)	0.797 (0.797)	0.914 (0.033)	0.672 (0.085)	0.762 (0.066)	0.822 (0.051)	0.017 (0.003)
			3	0.850 (0.039)	0.830 (0.830)	0.921 (0.028)	0.693 (0.080)	0.819 (0.048)	0.843 (0.040)	0.018 (0.017)
		MERSS	5	0.855 (0.037)	0.857 (0.857)	0.933 (0.026)	0.712 (0.074)	0.853 (0.038)	0.855 (0.037)	0.017 (0.004)
			10	0.864 (0.038)	0.887 (0.887)	0.938 (0.024)	0.714 (0.082)	0.886 (0.033)	0.855 (0.043)	0.017 (0.003)
	0.7	SRS	–	0.927 (0.024)	0.882 (0.882)	0.973 (0.016)	0.825 (0.058)	0.875 (0.043)	0.909 (0.033)	0.018 (0.004)
			3	0.913 (0.029)	0.905 (0.905)	0.974 (0.015)	0.827 (0.059)	0.907 (0.034)	0.913 (0.029)	0.018 (0.004)
		MERSS	5	0.915 (0.027)	0.925 (0.925)	0.975 (0.013)	0.826 (0.055)	0.927 (0.024)	0.911 (0.030)	0.018 (0.004)
			10	0.934 (0.027)	0.952 (0.952)	0.980 (0.018)	0.832 (0.070)	0.955 (0.019)	0.911 (0.042)	0.020 (0.022)
300	0.3	SRS	–	0.832 (0.031)	0.761 (0.761)	0.866 (0.036)	0.589 (0.075)	0.697 (0.059)	0.768 (0.046)	0.015 (0.003)
			3	0.822 (0.031)	0.776 (0.776)	0.882 (0.030)	0.615 (0.068)	0.750 (0.049)	0.799 (0.037)	0.014 (0.002)
		MERSS	5	0.826 (0.032)	0.801 (0.801)	0.894 (0.029)	0.640 (0.067)	0.782 (0.044)	0.815 (0.035)	0.014 (0.002)
			10	0.826 (0.032)	0.818 (0.818)	0.905 (0.025)	0.652 (0.063)	0.813 (0.036)	0.824 (0.032)	0.016 (0.019)
	0.5	SRS	–	0.866 (0.029)	0.811 (0.811)	0.911 (0.025)	0.676 (0.068)	0.765 (0.053)	0.820 (0.040)	0.018 (0.004)
			3	0.856 (0.029)	0.839 (0.839)	0.926 (0.022)	0.704 (0.060)	0.826 (0.038)	0.849 (0.031)	0.018 (0.004)
		MERSS	5	0.856 (0.030)	0.857 (0.857)	0.933 (0.021)	0.714 (0.060)	0.852 (0.033)	0.856 (0.031)	0.020 (0.018)
			10	0.864 (0.030)	0.878 (0.878)	0.943 (0.018)	0.717 (0.062)	0.886 (0.025)	0.855 (0.035)	0.018 (0.004)
	0.7	SRS	–	0.930 (0.019)	0.894 (0.894)	0.976 (0.011)	0.833 (0.046)	0.882 (0.035)	0.913 (0.026)	0.022 (0.005)
			3	0.917 (0.023)	0.908 (0.908)	0.976 (0.011)	0.833 (0.046)	0.910 (0.025)	0.916 (0.023)	0.020 (0.004)
		MERSS	5	0.921 (0.023)	0.930 (0.930)	0.979 (0.010)	0.838 (0.047)	0.932 (0.020)	0.918 (0.024)	0.020 (0.004)
			10	0.937 (0.021)	0.953 (0.953)	0.984 (0.008)	0.841 (0.052)	0.957 (0.015)	0.915 (0.032)	0.023 (0.028)

Note: Results are reported as Mean(SD) over $M = 200$ replications.

demonstrated in Figures [1a–1c](#).

Table 3: Performance metrics for logistic regression models under Imbalanced class condition ($r = 0.2$), evaluated across different sample sizes, varying set sizes, and correlation levels.

n	ρ	Method	k	Accuracy	Precision	Pr-AUC	MCC	Minor-F1	G-Mean	Run-Time
120	0.3	SRS	–	0.849 (0.045)	0.657 (0.657)	0.832 (0.067)	0.477 (0.152)	0.548 (0.137)	0.669 (0.122)	0.014 (0.003)
			3	0.820 (0.048)	0.688 (0.688)	0.850 (0.059)	0.519 (0.122)	0.626 (0.107)	0.726 (0.085)	0.014 (0.003)
		MERSS	5	0.816 (0.050)	0.725 (0.725)	0.858 (0.056)	0.553 (0.116)	0.671 (0.092)	0.752 (0.072)	0.014 (0.003)
			10	0.817 (0.052)	0.754 (0.754)	0.880 (0.054)	0.595 (0.117)	0.726 (0.085)	0.786 (0.065)	0.014 (0.003)
	0.5	SRS	–	0.871 (0.044)	0.713 (0.713)	0.889 (0.060)	0.592 (0.139)	0.659 (0.124)	0.765 (0.098)	0.015 (0.003)
			3	0.850 (0.049)	0.777 (0.777)	0.898 (0.053)	0.643 (0.124)	0.741 (0.097)	0.806 (0.073)	0.016 (0.003)
		MERSS	5	0.848 (0.049)	0.802 (0.802)	0.910 (0.044)	0.671 (0.107)	0.785 (0.076)	0.829 (0.058)	0.015 (0.003)
			10	0.847 (0.048)	0.836 (0.836)	0.925 (0.035)	0.696 (0.095)	0.838 (0.052)	0.846 (0.048)	0.015 (0.003)
	0.7	SRS	–	0.941 (0.033)	0.857 (0.857)	0.949 (0.048)	0.819 (0.103)	0.850 (0.088)	0.906 (0.065)	0.018 (0.004)
			3	0.922 (0.049)	0.878 (0.878)	0.951 (0.049)	0.830 (0.102)	0.885 (0.071)	0.915 (0.051)	0.018 (0.003)
		MERSS	5	0.926 (0.035)	0.922 (0.922)	0.961 (0.035)	0.852 (0.069)	0.917 (0.041)	0.925 (0.035)	0.018 (0.003)
			10	0.918 (0.039)	0.935 (0.935)	0.953 (0.038)	0.828 (0.080)	0.933 (0.032)	0.912 (0.044)	0.018 (0.004)
210	0.3	SRS	–	0.850 (0.037)	0.669 (0.669)	0.829 (0.054)	0.483 (0.118)	0.555 (0.109)	0.672 (0.089)	0.020 (0.004)
			3	0.835 (0.039)	0.719 (0.719)	0.860 (0.046)	0.552 (0.098)	0.652 (0.080)	0.742 (0.062)	0.020 (0.004)
		MERSS	5	0.828 (0.037)	0.746 (0.746)	0.867 (0.042)	0.578 (0.091)	0.689 (0.072)	0.765 (0.056)	0.019 (0.004)
			10	0.826 (0.039)	0.779 (0.779)	0.885 (0.040)	0.616 (0.086)	0.743 (0.063)	0.796 (0.049)	0.020 (0.004)
	0.5	SRS	–	0.888 (0.031)	0.763 (0.763)	0.903 (0.038)	0.633 (0.096)	0.692 (0.084)	0.780 (0.071)	0.017 (0.003)
			3	0.864 (0.037)	0.793 (0.793)	0.911 (0.034)	0.670 (0.083)	0.761 (0.062)	0.822 (0.051)	0.016 (0.004)
		MERSS	5	0.855 (0.034)	0.819 (0.819)	0.919 (0.030)	0.689 (0.074)	0.800 (0.051)	0.838 (0.042)	0.017 (0.004)
			10	0.852 (0.036)	0.853 (0.853)	0.929 (0.027)	0.706 (0.072)	0.844 (0.040)	0.851 (0.036)	0.016 (0.004)
	0.7	SRS	–	0.954 (0.021)	0.886 (0.886)	0.969 (0.032)	0.858 (0.068)	0.884 (0.057)	0.928 (0.042)	0.018 (0.003)
			3	0.941 (0.024)	0.918 (0.918)	0.976 (0.023)	0.870 (0.054)	0.914 (0.037)	0.934 (0.029)	0.018 (0.003)
		MERSS	5	0.938 (0.024)	0.931 (0.931)	0.981 (0.015)	0.874 (0.047)	0.930 (0.027)	0.937 (0.023)	0.018 (0.003)
			10	0.938 (0.025)	0.951 (0.951)	0.980 (0.022)	0.867 (0.053)	0.950 (0.020)	0.932 (0.028)	0.018 (0.003)
300	0.3	SRS	–	0.859 (0.030)	0.714 (0.714)	0.840 (0.044)	0.511 (0.091)	0.575 (0.085)	0.680 (0.070)	0.021 (0.005)
			3	0.844 (0.029)	0.752 (0.752)	0.869 (0.033)	0.578 (0.075)	0.670 (0.062)	0.751 (0.048)	0.021 (0.005)
		MERSS	5	0.836 (0.032)	0.761 (0.761)	0.877 (0.034)	0.598 (0.075)	0.706 (0.059)	0.776 (0.047)	0.022 (0.018)
			10	0.825 (0.032)	0.771 (0.771)	0.885 (0.030)	0.613 (0.068)	0.743 (0.048)	0.797 (0.038)	0.021 (0.004)
	0.5	SRS	–	0.890 (0.025)	0.769 (0.769)	0.907 (0.032)	0.644 (0.078)	0.705 (0.068)	0.789 (0.053)	0.019 (0.004)
			3	0.867 (0.029)	0.801 (0.801)	0.915 (0.028)	0.678 (0.070)	0.767 (0.055)	0.825 (0.042)	0.019 (0.004)
		MERSS	5	0.861 (0.027)	0.825 (0.825)	0.924 (0.023)	0.700 (0.058)	0.807 (0.039)	0.844 (0.030)	0.019 (0.003)
			10	0.856 (0.027)	0.854 (0.854)	0.933 (0.020)	0.712 (0.055)	0.848 (0.031)	0.855 (0.027)	0.019 (0.004)
	0.7	SRS	–	0.960 (0.017)	0.906 (0.906)	0.983 (0.020)	0.875 (0.054)	0.898 (0.045)	0.934 (0.032)	0.021 (0.004)
			3	0.947 (0.017)	0.925 (0.925)	0.987 (0.009)	0.883 (0.037)	0.922 (0.025)	0.941 (0.019)	0.020 (0.004)
		MERSS	5	0.941 (0.019)	0.932 (0.932)	0.987 (0.007)	0.881 (0.039)	0.934 (0.023)	0.940 (0.019)	0.020 (0.005)
			10	0.945 (0.018)	0.955 (0.955)	0.988 (0.008)	0.882 (0.038)	0.956 (0.015)	0.940 (0.021)	0.020 (0.005)

Note: Results are reported as Mean(SD) over $M = 200$ replications.

Discussion and Conclusion

This study evaluated the performance of MERSS for LR classification under imbalanced conditions. Our results demonstrated that MERSS consistently outperforms SRS, particularly for minority class identifica-

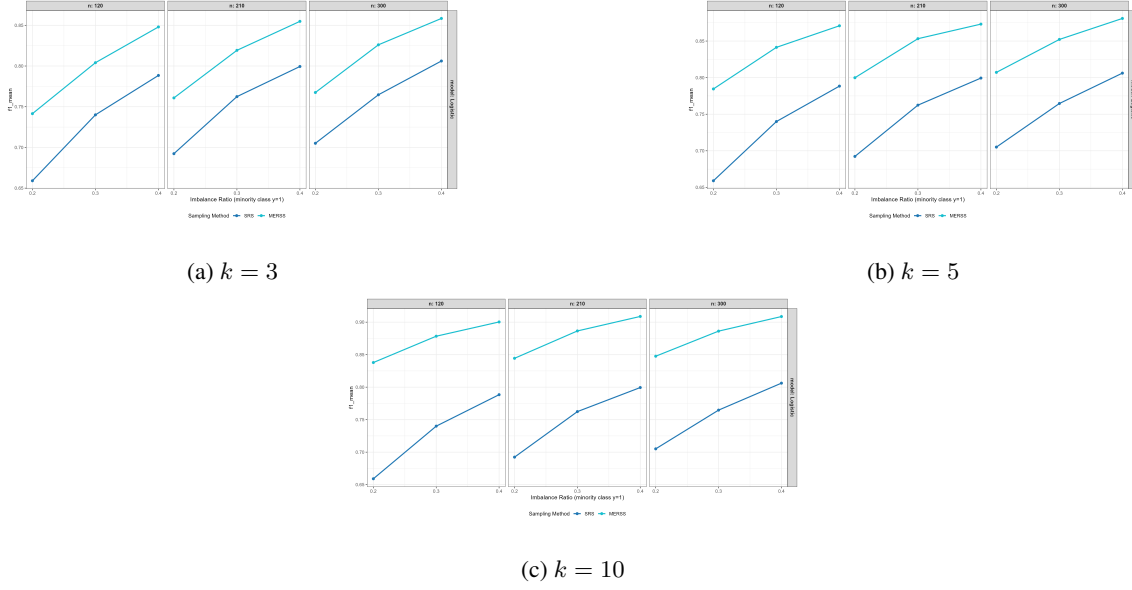


Figure 1: Minor-F1 score for MERSS vs. SRS at $\rho = 0.5$ across different sample sizes and set sizes.

tion. These findings align with previous research demonstrating the efficiency of RBS methods [Santos and Barrios \(2017\)](#); [Yousefi et al. \(2026\)](#) and extend the work of Samawi et al. [Samawi et al. \(2017, 2020\)](#) by focusing specifically on classification performance rather than parameter estimation.

The advantages of MERSS were most pronounced under severe imbalance ($r = 0.2$) and moderate to high correlation ($\rho = 0.5, 0.7$), where improvements in Precision and Minor-F1 reached up to 15% and 29%, respectively, compared to SRS. Increasing set size (k) from 3 to 10 consistently enhanced MERSS performance, consistent with Samawi et al. [Samawi et al. \(2017\)](#), [Santos and Barrios \(2017\)](#), and [Yousefi et al. \(2026\)](#). Notably, these gains were achieved with no additional computational cost, supporting the practical utility of MERSS in resource-limited settings.

On the other hand, unlike synthetic data methods such as SMOTE [Elreedy et al. \(2024\)](#), MERSS selects actual observational units using auxiliary information, preserving data authenticity while enriching minority class representation. This characteristic makes MERSS particularly valuable in different research where data integrity and inferential validity are paramount.

In conclusion, MERSS reliably enhances LR classification under imbalance, improving minority class identification at no additional computational cost. The method is most effective with moderate to strong auxiliary correlation and larger set sizes. These findings establish MERSS as a practical sampling strategy for rare event research with limited resources.

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